

What to make with excess solar?

In May 2019, California hit two records in regards to solar energy: (1) most solar ever flowing into the state's electrical grid, and (2) the most solar electricity ever taken offline because it wasn't needed¹. In fact, California produces so much solar energy, the state often needs to curtail production when they are not able to sell excess solar energy to nearby states².

Inspired by the unique problem of too much energy, and with an eye toward impacting carbon emissions, the venture firm XS Solar Ventures (XSSV) was formed to invest in ideas for storing, and creatively using the excess solar that would otherwise be wasted. As a previously successful entrepreneur and chemical engineer, XSSV asks if you would be interested in participating in a challenge, proposing potential ways of using or storing the excess energy chemically.

After you accept the offer from XSSV, you receive the following email:

From: XSSV Partners (partners@xssv.com)
Subject: XSSV Challenge II

Dear Colleagues,

Thank you for your interest in the XSSV Challenge II. We want to use this opportunity to elaborate on our vision and report on where we are now. To preface our challenge, we would like to share an excerpt from an announcement for a similar call for proposals put on by our colleagues at ARPA-E in the DOE:

We are living in challenging times, but are surrounded by opportunities. The widespread use of fossil fuels has long driven the engine of economic growth, and yet our dependence on these fuels severely threatens our national and environmental security due to our growing foreign energy dependence as well as climate change. Business as usual is not an option, as the outcome will be devastating. This is true not only for the U.S., but also for all nations in this interconnected world. The nation that successfully grows its economy with more efficient energy use, a clean domestic energy supply, and a smart energy infrastructure will lead the global economy of the 21st century. In many cases, we are lagging behind. We as a nation need to change course with fierce urgency.

Let us try to comprehend the scale of this challenge with the following question: What were those innovations of the 20th century that changed the course of humankind's history? Perhaps the most important one was the Haber-Bosch process of creating artificial fertilizers by fixing atmospheric nitrogen to form ammonia. It touched humanity like none other because it led to massive increase in food production and an almost four-fold increase in global population in 100 years. But there were plenty of other

game-changers: creating semi-dwarf, high-yield strains of wheat that introduced the green revolution; antibiotics; polio vaccination; the transistor and integrated circuits; electrification; the airplane; nuclear energy; optical and wireless communication; the internet; and so on. Now imagine all of these innovations happening in a span of just 10-20 years: That is the scale and pace of game-changing innovations that we now need to address the energy and climate change challenge of our and future generations. (3)

Our hope with XSSV's second challenge is to do what ARPA-E is within DOE - jumpstart a transformation in technology that will help address our climate change challenges. In our first fund, we were able to support a number of high profile companies who had used this excess solar to create a variety of different chemical products - from those used to store energy and those used for commercial and personal applications. Here, we wish to continue our purpose-driven fund with by requesting kernels of ideas of what chemical products could be targeted in the XSSV Challenge II. As our first challenge was so competitive, we are looking forward to those who weren't able to join in our first fund to reapply with their improved ideas.

Specifically we are looking for you and your teams to propose: (1) a use for the excess solar energy that (2) uses carbon dioxide to create a chemical product of interest (anything from chemical precursors to other products that could be made in high volume, or a specialty material that can be sold for a high margin). You can assume that the energy and carbon dioxide can be obtained for free when you propose your idea.

To move onto the second round, please submit a short slide deck (about 5 slides) and a 2-3 page write up that outlines your idea for the use of the excess solar and carbon dioxide with the following details:

1. A description chemical product you have decided to target
2. An explanation as to why this product is ideal to target (e.g. What are its properties and applications? What can we sell it for? Why is it in demand?)
3. The process diagram by which you plan to use the excess solar energy and carbon dioxide to form the target (e.g. chemical reactions, high level process flow diagram, highlighting the use of CO₂ and energy), include what material and/or energy would be going in and out of the process
4. An explanation as to why the process and product(s) you have targeted fits our mission in investing in green, profitable companies and is better than other choices we will see submitted in this challenge.

We are more than happy to answer any questions that you may have around the limitations of the challenge, but the XSSV partners are excited to see what creative solutions your teams will submit.

Sincerely,
XSSV General Partners

With the challenge message in hand, work with your team to propose a chemical product and production process, along with the chemical reactions to synthesize that product, materials and energy resources needed in and out of the process, and your reasoning for your team's selection. As always, you are free to look up any information you feel is necessary to help you solve the problem, making sure to cite your resources or references when applicable.

References

1. Roth, S. "California has too much solar power. That might be good for ratepayers," *Los Angeles Times*. June 5, 2019.
2. Cotting, A, & Hering, G. "Solar-fueled oversupplies drive record renewable energy waste in California," *S&P Global Market Intelligence*. June 10, 2019.
3. US Department of Energy Headquarters Advanced Research Projects Agency – Energy (ARPA-E), "Electrofuels," *Recovery Act, Financial Assistance Funding Opportunity Announcement*. 2010.